4.5.5 Pressure and pressure differences in fluids (physics only)

4.5.5.1 Pressure in a fluid

4.5.5.1.1 Pressure in a fluid 1

Content	Key opportunities for skills development
A fluid can be either a liquid or a gas. The pressure in fluids causes a force normal (at right angles) to any surface. The pressure at the surface of a fluid can be calculated using the equation:	

Content	Key opportunities for skills development
pressure = $\frac{\text{force normal to a surface}}{\text{area of that surface}}$ $\left[p = \frac{F}{A}\right]$ pressure, p, in pascals, Pa force, F, in newtons, N area, A, in metres squared, m^2	MS 3b, c WS 4.3, 4.4, 4.5, 4.6 Students should be able to recall and apply this equation.

4.5.5.1.2 Pressure in a fluid 2 (HT only)

Content	Key opportunities for skills development
The pressure due to a column of liquid can be calculated using the equation:	
pressure = height of the column × density of the liquid × gravitational field strength [$p = h \rho g$] pressure, p, in pascals, Pa height of the column, h, in metres, m density, ρ, in kilograms per metre cubed, kg/m^3 gravitational field strength, g, in newtons per kilogram, N/kg (In any calculation the value of the gravitational field strength (g) will be given.) Students should be able to explain why, in a liquid, pressure at a point increases with the height of the column of liquid above that point and with the density of the liquid.	MS 3b, 3c WS 4.3, 4.4, 4.5, 4.6 Students should be able to apply this equation which is given on the Physics equation sheet.
Students should be able to calculate the differences in pressure at different depths in a liquid. A partially (or totally) submerged object experiences a greater pressure on the bottom surface than on the top surface. This creates a resultant force upwards. This force is called the upthrust. Students should be able to describe the factors which influence floating and sinking.	MS 1c, 3c

4.5.5.2 Atmospheric pressure

Content	Key opportunities for skills development
The atmosphere is a thin layer (relative to the size of the Earth) of air round the Earth. The atmosphere gets less dense with increasing altitude. Air molecules colliding with a surface create atmospheric pressure. The number of air molecules (and so the weight of air) above a surface decreases as the height of the surface above ground level increases. So as height increases there is always less air above a surface than there is at a lower height. So atmospheric pressure decreases with an increase in height. Students should be able to:	
• describe a simple model of the Earth's atmosphere and of atmospheric pressure • explain why atmospheric pressure varies with height above a surface.	WS 1.2